

Confusion Matrix — GRU-AE (Th = 0.014309, FPR(Val) ≤ 5%)

Rule-Only

Actual	Predicted	
	Normal	Anomaly
Normal	5,674	49
Anomaly	318	1,918

Recall 85.78% F1 91.27%
FPR(Attack) 0.86% FPR(Val) 2.93%

ML-Only

Actual	Predicted	
	Normal	Anomaly
Normal	5,613	110
Anomaly	97	2,139

Recall 95.66% F1 95.38%
FPR(Attack) 1.92% FPR(Val) 2.99%

Hybrid

Actual	Predicted	
	Normal	Anomaly
Normal	5,605	118
Anomaly	43	2,193

Recall 98.08% F1 96.46%
FPR(Attack) 2.06% FPR(Val) 4.97%